# STREET FIGHTER AR:
## A FIRST-PERSON AR FITNESS APP

CG4002 CAPSTONE PROJECT | GROUP 19
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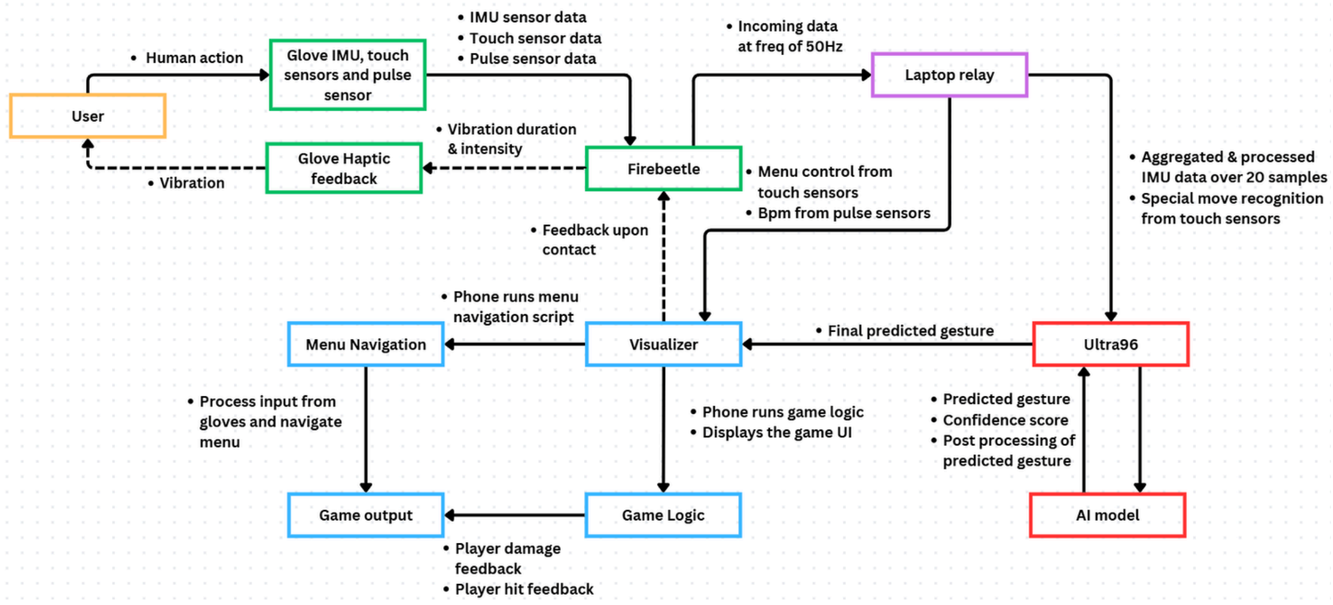
## 1 | MOTIVATION & USE CASES

- **The Problem:** The rise of sedentary lifestyles and prolonged screen time.
- **The Solution:** A multiplayer first-person AR fighting game. Physical movement replace button presses, encouraging cardiovascular exercise.
- **Use Cases:** Attack and defend in real time, use special abilities, and scale in-game stamina based on real-world heart rate.




## 2 | SYSTEM ARCHITECTURE

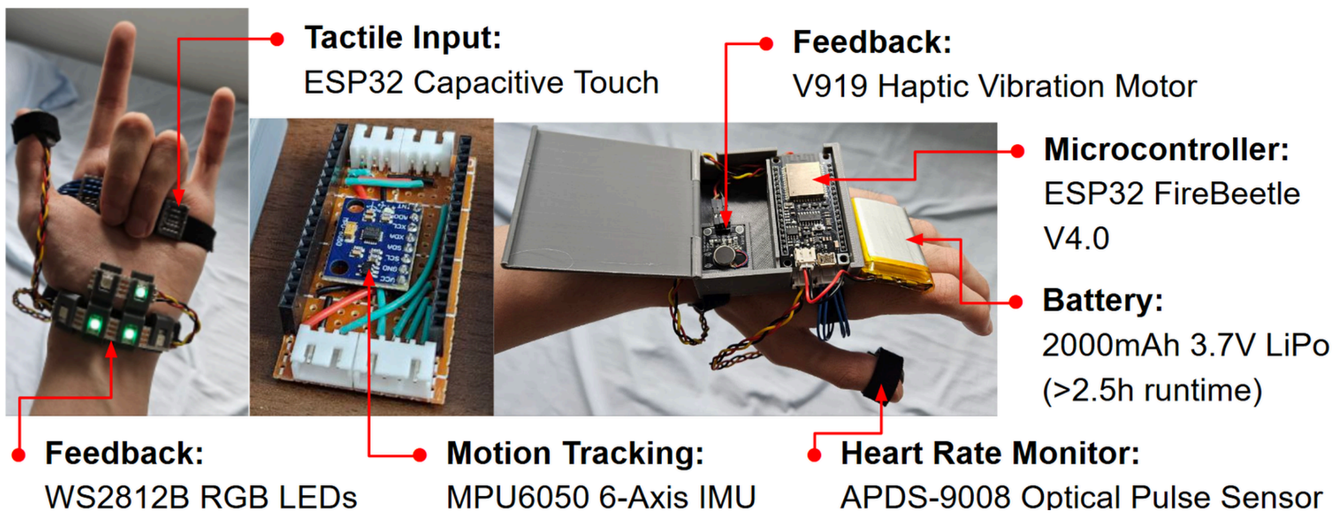
- Sensor Nodes: Wearable Gloves (ESP32 and IMU) capture motion & biometrics.
- Network Hub: Mosquitto MQTT Broker hosted on AWS.
- Brain: Ultra96 FPGA processes data through 1D-CNN to classify gestures.
- Display: Android Phone renders the AR environment.

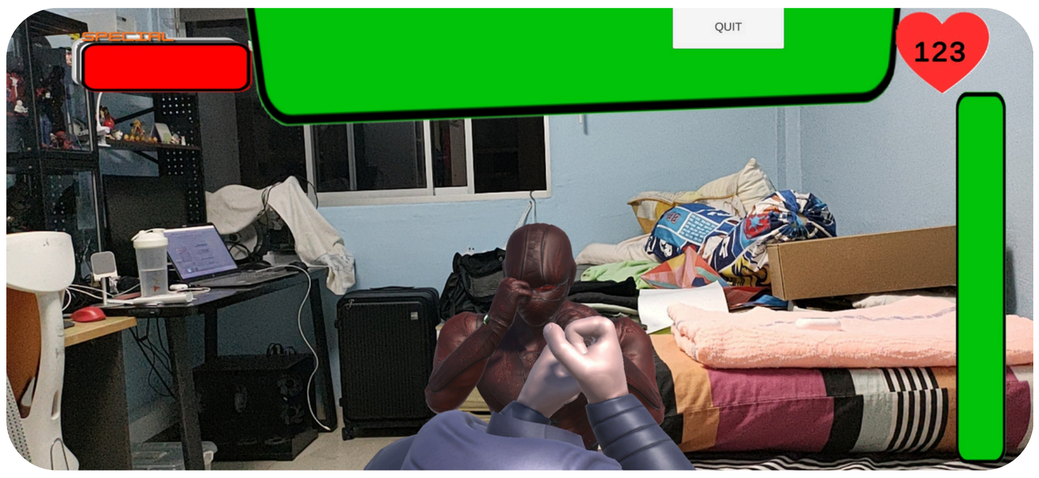
## 3 | COMMUNICATIONS

- Protocol: MQTT over Wi-Fi for lightweight, real-time telemetry.
- Security: **TLS encryption** with a Certificate-Based Trust model prevents interception. **Access control** implemented by requiring valid username & password
- Broker: Mosquitto hosted through AWS for **remote connection**.
- Concurrency: Topic-based filtering, threading implemented at Laptop and Ultra96 to **handle data stream from multiple players**

## 4 | GLOVE HARDWARE

- **50Hz Control Loop**
  - Deterministic data acquisition optimized for efficiency.
  - Zero-latency gesture recognition and haptic feedback.
- **Advanced Signal Filtering**
  - Exponential Moving Average (EMA) filter for clear ECG.
  - Missed-beat interpolator to maintain consistent BPM.
- **Ergonomic & Modular**
  - Custom 3D-printed casing protects electronics.
  - Shrouded pulse sensor to block ambient noise.




## 5 | EDGE AI PROCESSING (ULTRA96)

- 1D-CNN Architecture: Classifies 10 combat moves by extracting the features from a 100ms rolling window of 6-axis IMU data, processed as a 6 channel x 20 sample input tensor.
- Average total end-to-end latency of 37μs.
- Average model latency of the model is 36.28 μs at 100MHz.
- High bandwidth data movement: Utilises the AXI4-Master (M_AXI) interface to perform burst transfers from the DRAM to the FPGA BRAM. This bypasses the CPU data path, minimising CPU overhead, 1.9% of total latency.
- Model accuracy of >80% during real time gaming

## 6 | AR VISUALIZER (UNITY)

- **Third-Person Immersion:** Built on Unity AR Foundation & Google ARCore.
- **Dynamic UI**: Real-time health bars, special ability meters and sprites anchored to the AR space.
- **Responsive Feedback:** MQTT moves triggers effects, character animations, and visual damage perfectly synced with the glove's haptic motor and phone screens.
- **Multiplayer Sync:** Niantic Lightship simplifies device-to-device spatial alignment for local PvP fights. Netcode is used for multiplayer networking.